Product Summary

$V_{(BR)DSS}$	$R_{DS(ON)}$ Max	Package	I_D T_A = +25°C
-20V	52mΩ @V_{GS} = -4.5V	SOT23	-5.0A
	100mΩ @V_{GS} = -2.5V		-3.6A

Description

This MOSFET is designed to minimize the on-state resistance ($R_{DS(ON)}$), yet maintain superior switching performance, making it ideal for high efficiency power management applications.

Features

- Low On-Resistance
- Low Input Capacitance
- Fast Switching Speed
- **Totally Lead-Free & Fully RoHS Compliant (Notes 1 & 2)**
- **Halogen and Antimony Free. "Green" Device (Note 3)**
- **Qualified to AEC-Q101 standards for High Reliability**
- **An Automotive-Compliant Part is Available Under Separate Datasheet (DMG230UXQ)**

Applications

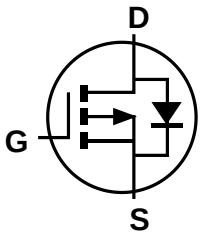
- Backlighting
- Power Management Functions
- DC-DC Converters
- Motor Control

Mechanical Data

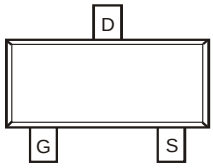
- Case: SOT23
- Case Material: Molded Plastic, "Green" Molding Compound. UL Flammability Classification Rating 94V-0
- Moisture Sensitivity: Level 1 per J-STD-020D
- Terminals: Finish – Matte Tin Annealed over Copper Leadframe. Solderable per MIL-STD-202, Method 208 e3
- Terminals Connections: See Diagram Below
- Weight: 0.008 grams (Approximate)



Top View



Internal Schematic



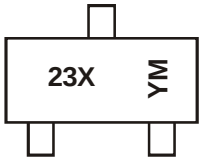
Top View

Ordering Information (Note 4)

Part Number	Compliance	Case	Packaging
DMG2305UX-7	Standard	SOT23	3,000/Tape & Reel
DMG2305UX-13	Standard	SOT23	10,000/Tape & Reel

Notes:
1. No purposely added lead. Fully EU Directive 2002/95/EC (RoHS) & 2011/65/EU (RoHS 2) compliant.
2. See http://www.diodes.com/quality/lead_free.html for more information about Diodes Incorporated's definitions of Halogen- and Antimony-free, "Green" and Lead-free.
3. Halogen- and Antimony-free "Green" products are defined as those which contain <900ppm bromine, <900ppm chlorine (<1500ppm total Br + Cl) and <1000ppm antimony compounds.
4. For packaging details, go to our website at http://www.diodes.com/products/packages.html.

Marking Information



23X = Product Type Marking Code
YM = Date Code Marking
Y or $\overline{Y}$= Year (ex: D = 2016)
M = Month (ex: 9 = September)

Date Code Key

Year	2009	~	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Code	W	~	D	E	F	G	H	I	J	K	L	M	N

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Code	1	2	3	4	5	6	7	8	9	O	N	D

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Maximum Ratings (@T_A = +25°C, unless otherwise specified.)

Characteristic			Symbol	Value	Units
Drain-Source Voltage			V_{DSS}	-20	V
Gate-Source Voltage			V_{GSS}	±8	V
Continuous Drain Current (Note 5) V_{GS} = -4.5V	Steady State	T_A = +25°C T_A = +70°C	I_D	-4.2 -3.3	A
	t<10s	T_A = +25°C T_A = +70°C	I_D	-5.0 -4.0	A
Pulsed Drain Current (Note 6)			I_{DM}	-10	A

Thermal Characteristics

Characteristic		Symbol	Value	Unit
Power Dissipation (Note 5)		P_D	1.4	W
Thermal Resistance, Junction to Ambient (Note 5)	Steady State	$R_{\theta JA}$	90	°C/W
	t<10s		64	°C/W
Thermal Resistance, Junction to Case (Note 7)		$R_{\theta JC}$	33	°C/W
Operating and Storage Temperature Range		T_J, T_{STG}	-55 to +150	°C

Electrical Characteristics (@T_A = +25°C, unless otherwise specified.)

Characteristic	Symbol	Min	Typ	Max	Unit	Test Condition
OFF CHARACTERISTICS (Note 7)						
Drain-Source Breakdown Voltage	BV_{DSS}	-20	—	—	V	V_{GS} = 0V, I_D = -250µA
Zero Gate Voltage Drain Current T_J = +25°C	I_{DSS}	—	—	-1.0	µA	V_{DS} = -20V, V_{GS} = 0V
Gate-Source Leakage	I_{GSS}	—	—	±100	nA	V_{GS} = ±8V, V_{DS} = 0V
ON CHARACTERISTICS (Note 7)						
Gate Threshold Voltage	$V_{GS(TH)}$	-0.5	—	-0.9	V	V_{DS} = V_{GS}, I_D = -250µA
Static Drain-Source On-Resistance	$R_{DS(ON)}$	—	40	52	mΩ	V_{GS} = -4.5V, I_D = -4.2A
			52	100		V_{GS} = -2.5V, I_D = -3.4A
			68	200		V_{GS} = -1.8V, I_D = -2A
Forward Transfer Admittance	$\|Y_{FS}\|$	—	9	—	S	V_{DS} = -5V, I_D = -4A
DYNAMIC CHARACTERISTICS (Note 8)						
Input Capacitance	C_{ISS}	—	808	—	pF	V_{DS} = -15V, V_{GS} = 0V f = 1.0MHz
Output Capacitance	C_{OSS}	—	85	—	pF	
Reverse Transfer Capacitance	C_{RSS}	—	77	—	pF	
Gate Resistance	R_G		15.2		Ω	V_{GS} = 0V, V_{DS} = 0V, f = 1.0MHz
SWITCHING CHARACTERISTICS (Note 8)						
Total Gate Charge	Q_G	—	10.2	—	nC	V_{GS} = -4.5V, V_{DS} = -4V, I_D = -3.5A
Gate-Source Charge	Q_{GS}	—	1.3	—	nC	
Gate-Drain Charge	Q_{GD}	—	2.2	—	nC	
Turn-On Delay Time	$t_{D(ON)}$	—	10.8	—	ns	V_{DS} = -4V, V_{GS} = -4.5V, R_G = 6Ω, I_D = -1A
Turn-On Rise Time	t_R	—	13.7	—	ns	
Turn-Off Delay Time	$t_{D(OFF)}$	—	79.3	—	ns	
Turn-Off Fall Time	t_F	—	34.7	—	ns	

Notes: 5. Device mounted on FR-4 substrate PC board, 2oz copper, with 1-inch square copper plate.
6. Repetitive rating, pulse width limited by junction temperature.
7. Short duration pulse test used to minimize self-heating effect.
8. Guaranteed by design. Not subject to product testing

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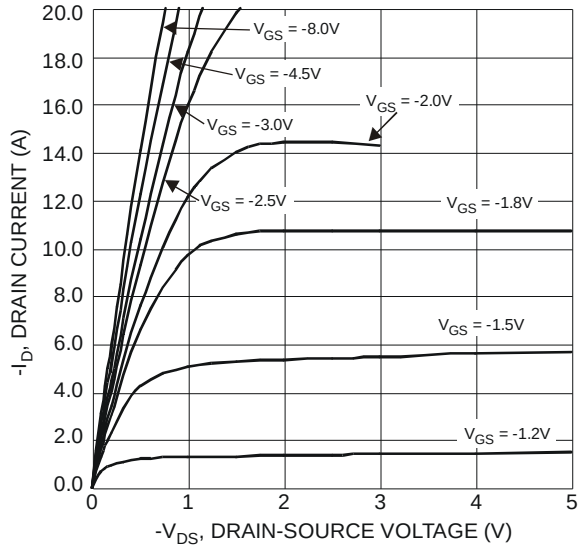


Figure 1 Typical Output Characteristics

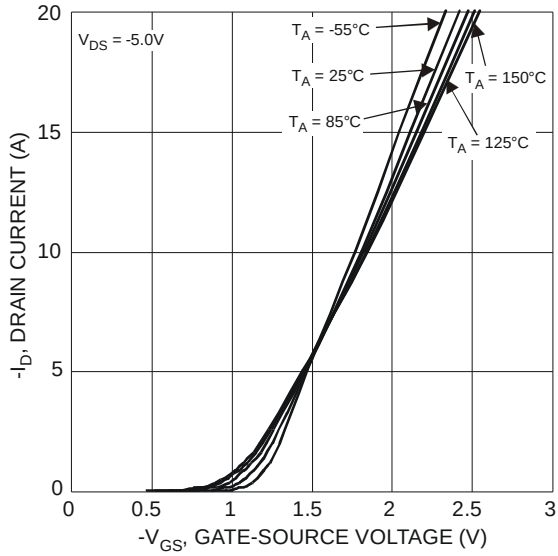


Figure 2 Typical Transfer Characteristics

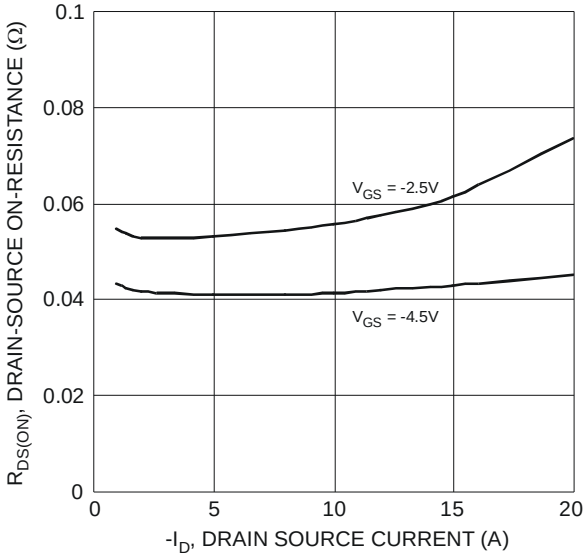


Figure 3 Typical On-Resistance vs. Drain Current and Gate Voltage

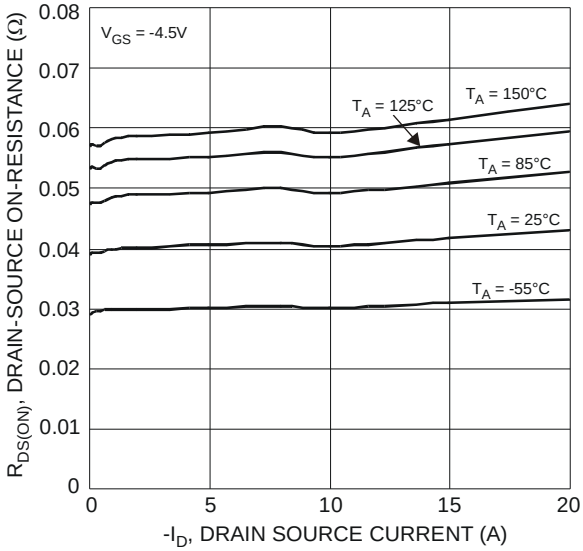


Figure 4 Typical On-Resistance vs. Drain Current and Temperature

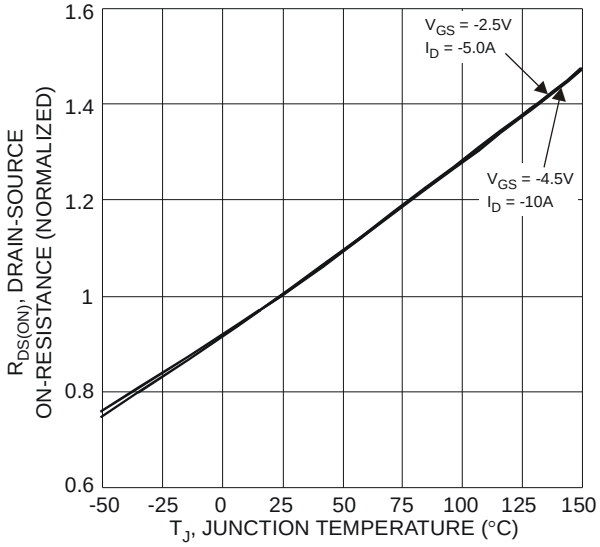


Figure 5 On-Resistance Variation with Temperature

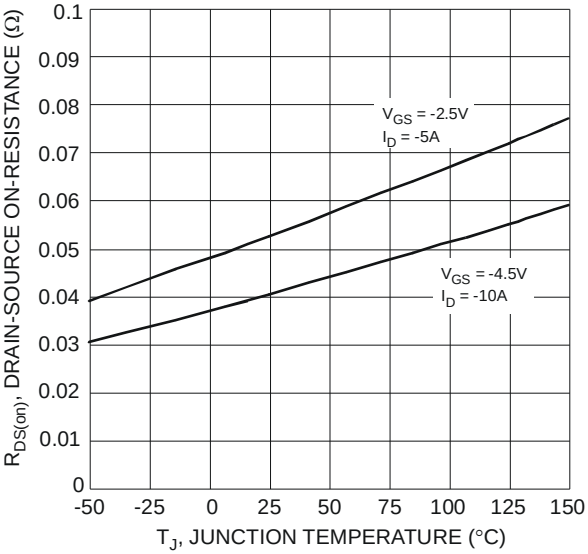


Figure 6 On-Resistance Variation with Temperature

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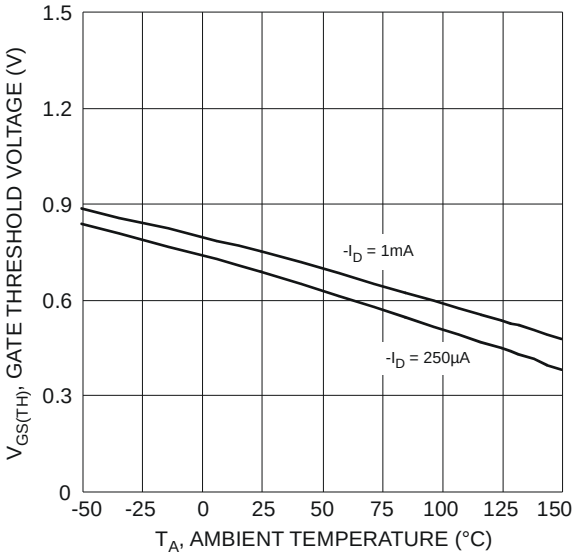


Figure 7 Gate Threshold Variation vs. Ambient Temperature

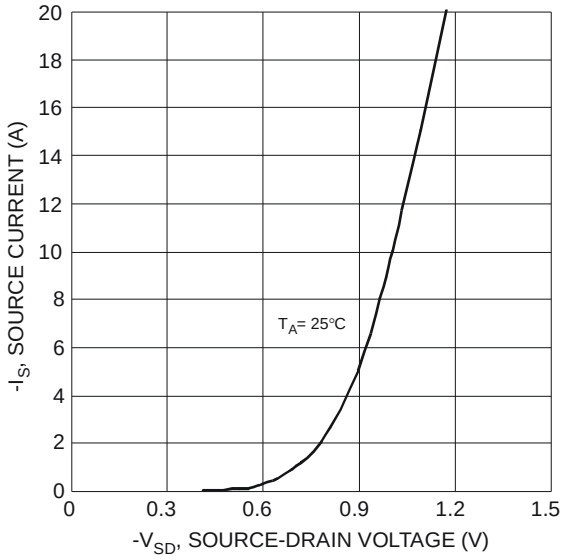


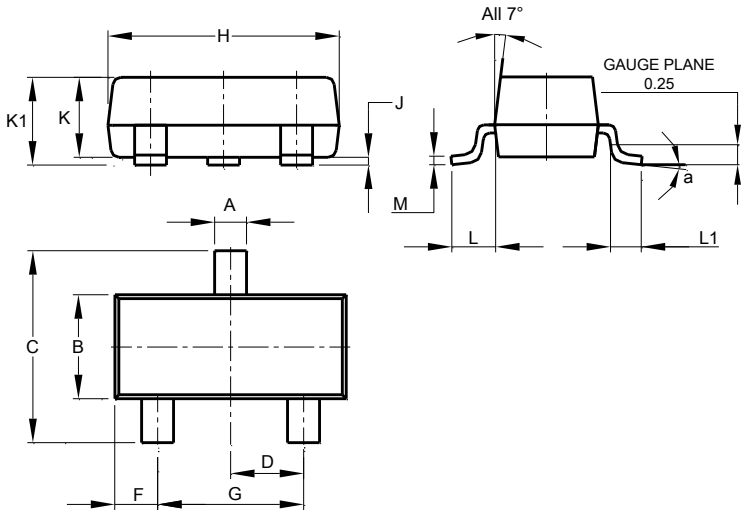
Figure 8 Diode Forward Voltage vs. Current

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Package Outline Dimensions

Please see http://www.diodes.com/package-outlines.html for the latest version.

SOT23

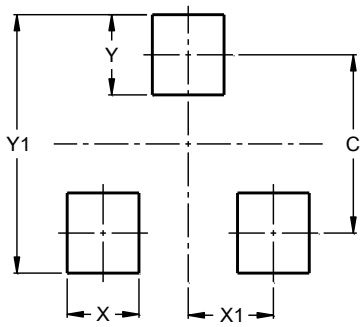


SOT23			
Dim	**Min**	**Max**	**Typ**
A	0.37	0.51	0.40
B	1.20	1.40	1.30
C	2.30	2.50	2.40
D	0.89	1.03	0.915
F	0.45	0.60	0.535
G	1.78	2.05	1.83
H	2.80	3.00	2.90
J	0.013	0.10	0.05
K	0.890	1.00	0.975
K1	0.903	1.10	1.025
L	0.45	0.61	0.55
L1	0.25	0.55	0.40
M	0.085	0.150	0.110
a	0°	8°	–
All Dimensions in mm			

Suggested Pad Layout

Please see http://www.diodes.com/package-outlines.html for the latest version.

SOT23



Dimensions	Value (in mm)
C	2.0
X	0.8
X1	1.35
Y	0.9
Y1	2.9

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